COMPUTER SCIENCE PROJECT

- Event Waitlist Management System -

CS 4485 – Team 12

Team members

- Disha Shetty - Frontend
- Angela Lopez - Frontend
- Neel Vavilapalli - Backend
- Kyle Albuquerque – Backend

Problem

- Large events still rely on manual or paper-based waitlists
- Existing digital tools are designed for small, single-venue settings
- These tools fail under high volume, multi-venue, low-connectivity conditions
- Results in inefficient capacity use, high staff workload, and poor attendee experience

Project Overview

Our project is a real-time event waitlist management platform designed to streamline entry management and room capacity across high-volume, multi-venue events with limited connectivity. It supports automated waitlist promotion, attendee notifications, and staff dashboards, and leverages AI to predict wait times and attendance, optimizing flow and improving the overall customer experience.

Key Features

QR-BASED, REAL-TIME WAITLISTS TIED TO ATTENDEE DATABASE

ROOM-LEVEL STAFF DASHBOARDS FOR CAPACITY & ENTRY MANAGEMENT

AUTOMATIC PROMOTION FROM WAITLIST ON CANCELLATIONS (SEAT/TABLE AWARE)

SMS/PUSH NOTIFICATIONS WITH TIME-LIMITED CHECK-IN

MULTI-VENUE, OFFLINE-FIRST SYSTEM WITH AI-BASED WAIT-TIME & NO-SHOW PREDICTION

Proposed Tech Stack

Frontend: React / Next.js (attendee UI + staff dashboard)

Backend: FastAPI (Python) or Node.js (REST APIs)

Database: PostgreSQL (waitlists, venues, users)